

Total No. of Questions : 4]

SEAT No. :

P8469

[Total No. of Pages : 2

Oct-22/BE/Insem-47

B.E. (Computer Engineering)

OBJECT ORIENTED MODELING & DESIGN

(2019 Pattern) (Semester - VII) (410244D) (Elective - III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain element of a class diagram with examples. [6]

b) What do you mean by Association class? [4]

c) Explain different Relations in UML. [5]

OR

Q2) a) What are different type of analysis classes? Explain with an example. [6]

b) How you model class & object in UML? [4]

c) Draw a class diagram for ATM system. [5]

Q3) a) Explain following : [5]

i) Composite state

ii) Self transition orthogonal states with respect to state chart diagram.

b) Draw state diagram with composite states from ATM card reading & authentication system. [5]

P.T.O.

- c) Explain terms : [5]
- i) Aggregation
 - ii) Composition

OR

- Q4) a) Explain different components of a state diagram. [5]
- b) Explain how you model different events in state diagram. [5]
- c) Draw state chart diagram for Railway Reservation system. [5]



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Aug - 2023

Total No. of Questions : 4]

SEAT No. :

P5195

[Total No. of Pages : 1

[6188]-147

B.E. (Computer Engineering) (Insem)
OBJECT ORIENTED MODELLING & DESIGN
(2019 Pattern) (Semester - VII) (Elective - III) (410244D)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain the types of relationship along with their notations. [5]
b) Draw class diagram for ATM system. [5]
c) Define object? Explain components of object diagram in brief? [5]

OR

- Q2)** a) Difference between link and association with suitable example. [5]
b) What is OO development? Explain the OO the mes ? [5]
c) Explain the following terms with respect to object oriented methodology. [5]
i) Abstraction
ii) Encapsulation
iii) Sharing

- Q3)** a) Draw state diagram with composite states from ATM card reading & authentication system. [5]
b) Write a short note on : [5]
i) Package
ii) Derived data
iii) Constraints
c) Define event and explain in brief different types of events. [5]

OR

- Q4)** a) Draw state chart diagram for Railway reservation system. [5]
b) What do you mean by Association ends? Explain N-ary association in short. [5]
c) What is state? Explain with example transitions and condition. [5]



Total No. of Questions : 4]

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SEAT No. :

[6361]-47

[Total No. of Pages : 2

B.E. (Computer Engineering) (Insem)
OBJECT ORIENTED MODELING & DESIGN
(2019 Pattern) (Semester- VII) (Elective - III) (410244D)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume the suitable data, if necessary.

- Q1) a) Draw a class diagram for following consider university database that keeps track of students & their majors, transcripts and registration & university courses several sections of each courses are offered and each section is related to instructor who is teaching. It also keep track of sponsored research Project of faculties & graduate students of academic departments of a particular college. The database also keep track of acknowledgement of each person & contacts awarded to each faculty. [6]
- b) What do you mean by abstract class? Explain with example. [4]
- c) Draw a class diagram for Railway Reservation system. [5]

OR

- Q2) a) Explain different methods of identifying analysis classes. [6]
- b) Differentiate between class diagram and object diagram in UML. [4]
- c) Explain concept of association class with an example. [5]
- Q3) a) Construct state transition diagram for computerization of your college library. [5]
- b) Explain different pseudo states in chart diagram. [5]
- c) Differentiate between terms Aggregation & Composition. [5]

OR

P.T.O.

Q2. Q4
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- Q4) a) Explain different components of a state diagram. [5]
b) Explain different types of states in state chart diagram. [5]
c) Draw state chart diagram for handling telecommunication system. [5]

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[Total No. of pages:1]

SEAT NO :

**Fourth Year of Computer Engineering
Object oriented Modeling and Design
(2019 Course) (410244(D)) (Semester – I)**

Time: 1.00 Hours

Max Marks: 30

Instructions to the candidates:

1. Attempt Q.1 or Q.2, Q.3 or 4
2. Neat diagrams must be drawn whenever necessary
3. Assume suitable data if necessary

- Q.1.a) Explain element of a class diagram with examples [5]
b) Explain the difference between link & association with suitable class. [5]
c) Explain the types of relationship along with their notations. [5]

OR

- Q.2.a) What are the different types of analysis classes? Explain with example [5]
b) Draw a class diagram for ATM system [5]
c) Explain the following terms with respect to object oriented methodology [5]

- Abstraction
- Encapsulation

- Q.3.a) Draw the state diagram for ATM card reading & authentication system [5]
b) Write a short note on [5]

- Package
- Derived data
- Constraints

- c) Define Event & Explain in brief different types of events. [5]

OR

- Q.4 a) Draw the state chart diagram for railway reservation system [5]
b) What is state? Explain with example transaction & condition [5]
c) Differentiate between procedure and object oriented language [5]